(I) Composite attributes are put as simplest attributes in relational model

(II) For multi-valued attributes, we have a separate table [PK → attribute + PK of original table]

(III) For weak-entity sets, combined attribute of its partial key and primary key of the strong-entity set it is dependent on, acts as PK of that table.

(IV) For 1:1 relations, the part having full participation includes the PK of the other part and any attribute of the relation is also added.

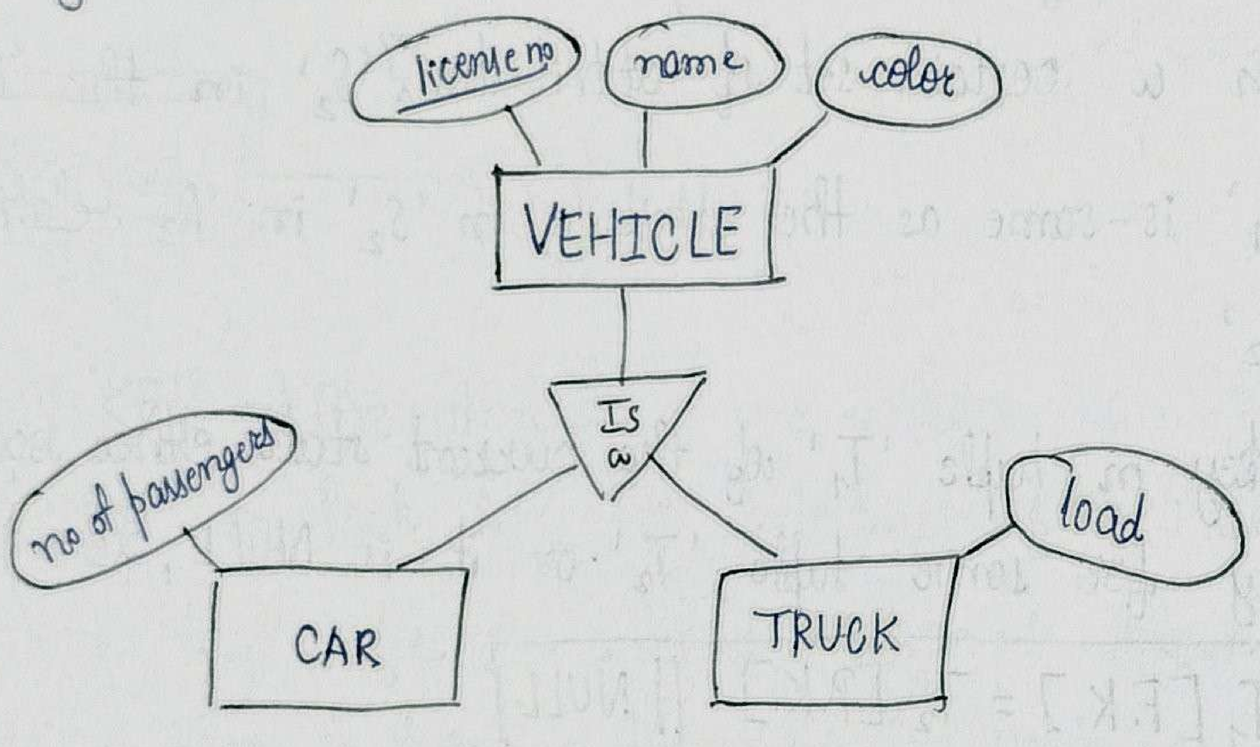
(V) For 1:1 relations, if both sides are full participation, both tables are merged with PK = combined PKs of both tables

(VI) For 1:N relation, the side with 'N' includes the P.K of the side with '1' as F.K.

(VII) For M:N relation, a table is created with P.K = combined P.Ks of both table.

(VIII) For self-referential F.K, a self-referential F.K. is added to that table
(1:N)

(IX) For ternary relation, a table is created with P.K = P.Ks of all three entities' P combi

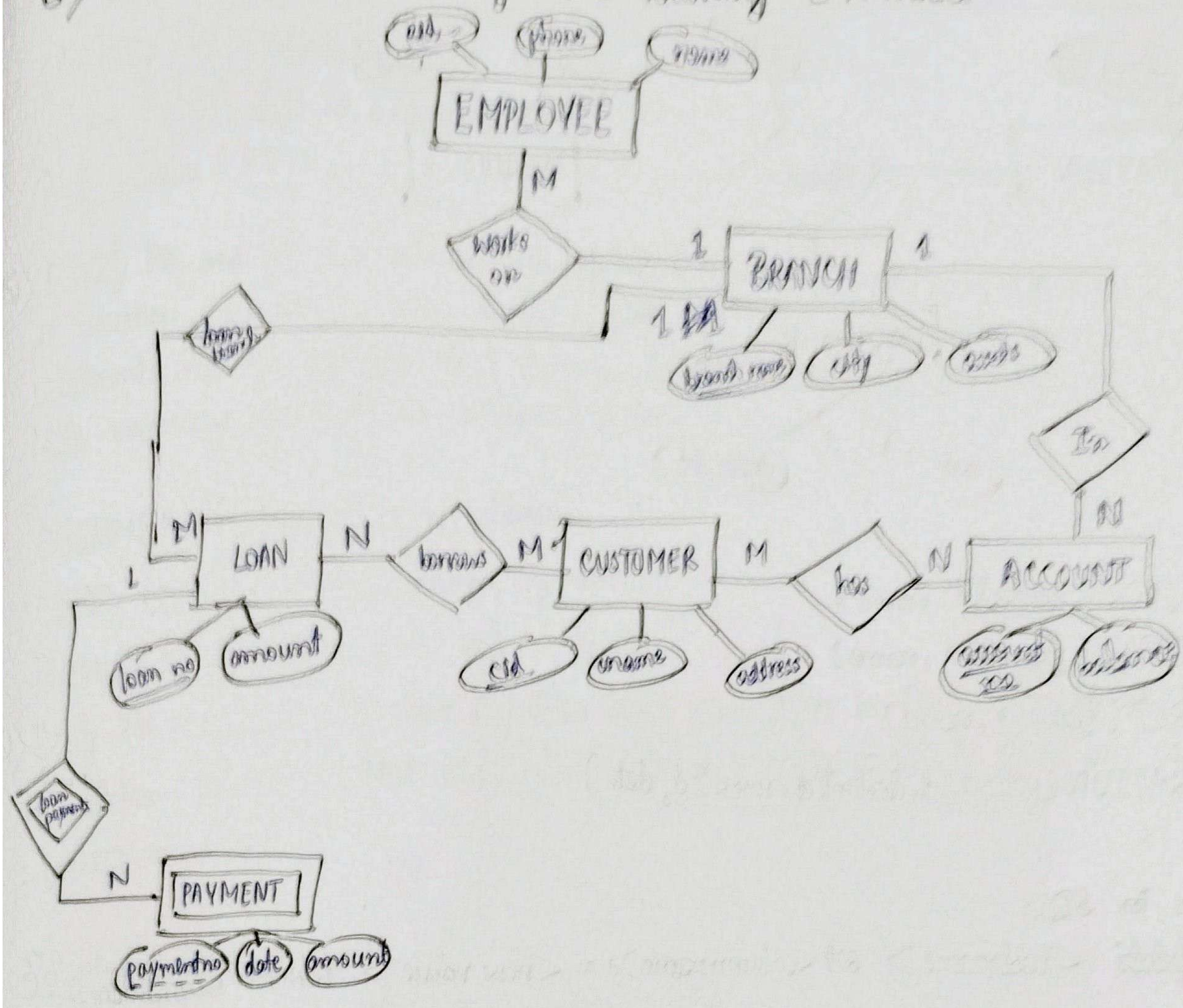


VEHICLE (license no., name, color)

CAR (license no., name, color, no. of passengers)

TRUCK (license no., name, color, load)

Q.) Draw relational model for the following E-R model



EMPLOYEE (eid, phone, name, branch name)

BRANCH (branch name, city, assets) ~~loan no~~

ACCOUNT(account no, balance, branch name)

CUSTOMER (cid, cname, address)

LOAN (loan no, amount, branch name)

PAYMENT (payment no, loan no, date, amount)

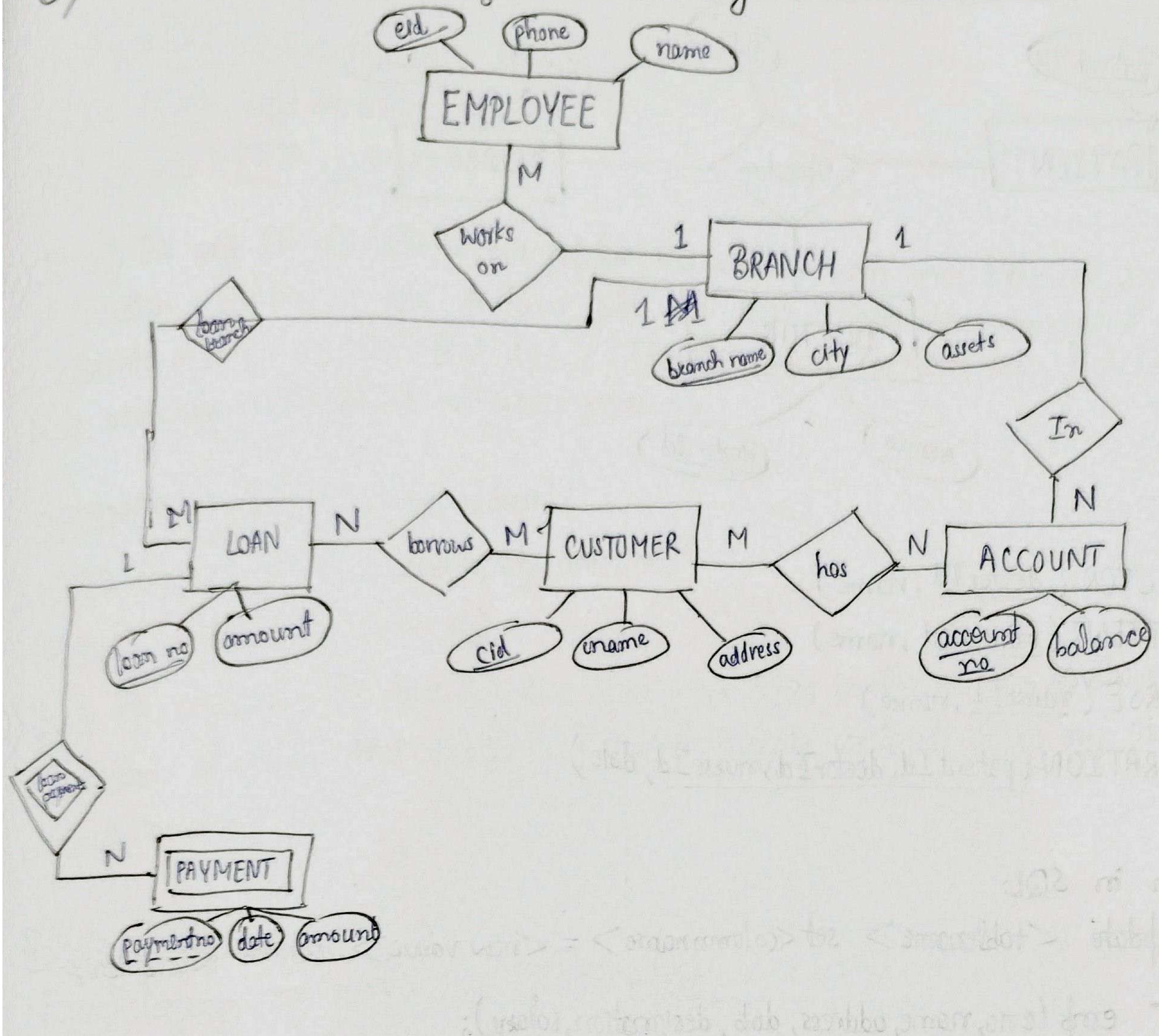
BORROW (cid, loan no)

DEPOSIT ~~HAS~~ (cid, account no)

or

CUSTOMER – ACCOUNT

8.) Draw relational model for the following E-R model



EMPLOYEE (eid, phone, name, branch name)

BRANCH (branch name, city, assets) ~~loan no~~

ACCOUNT (account no, balance, branch name)

CUSTOMER (Cid, cname, address)

LOAN (loan no, amount, branch name)

PAYMENT (payment no, loan no, date, amount)

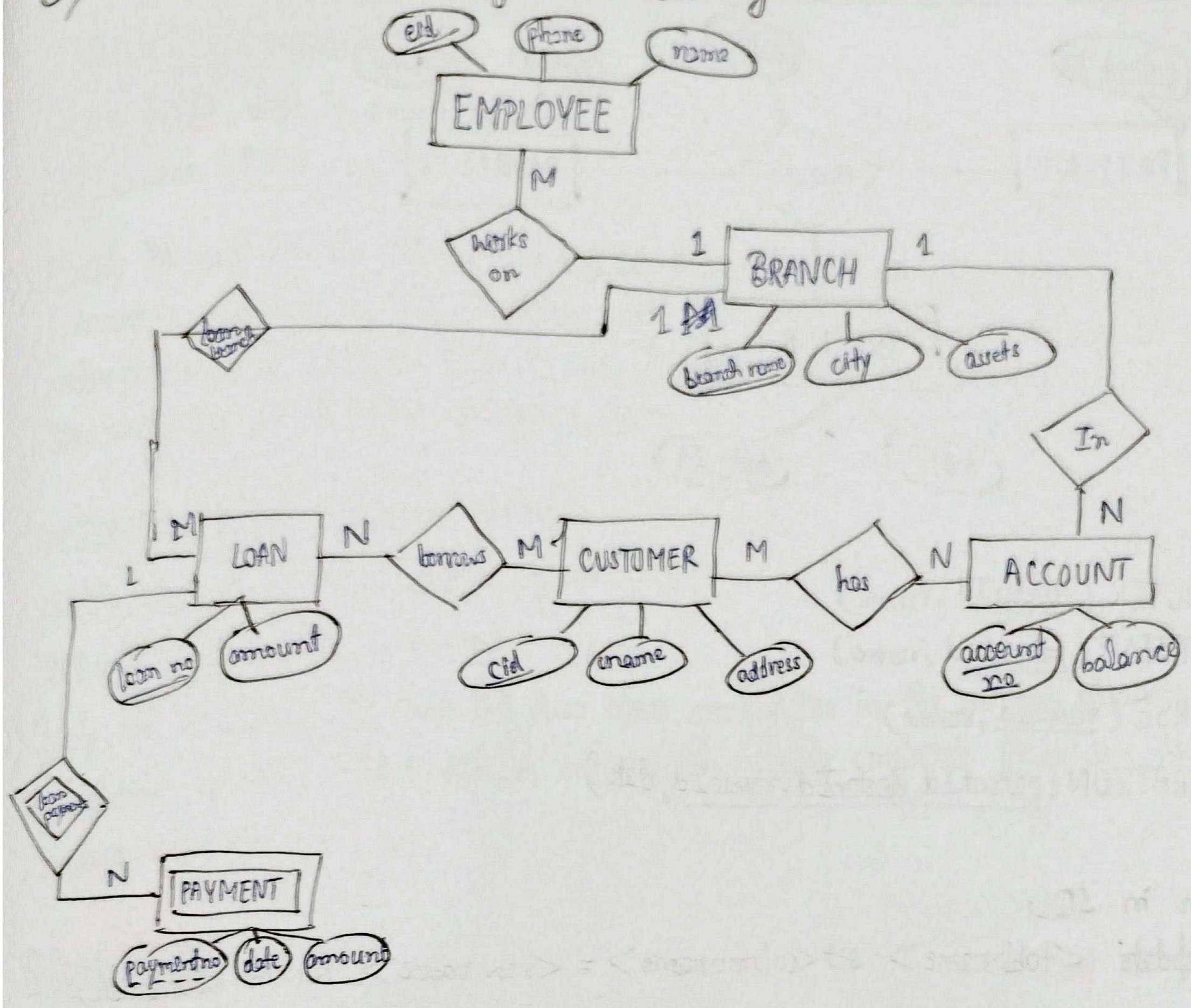
BORROW (cid, loan no)

DEPOSIT ~~HAS~~ (cid, account no)

or

CUSTOMER — ACCOUNT

8.) Draw relational model for the following E-R model



EMPLOYEE (eid, phone, name, branch name)

BRANCH (branch name, city, assets) ~~loan no~~

ACCOUNT(account no, balance, branch name)

CUSTOMER (cid, cname, address)

LOAN (loan no, amount, branch name)

PAYMENT (payment no, loan no, date, amount)

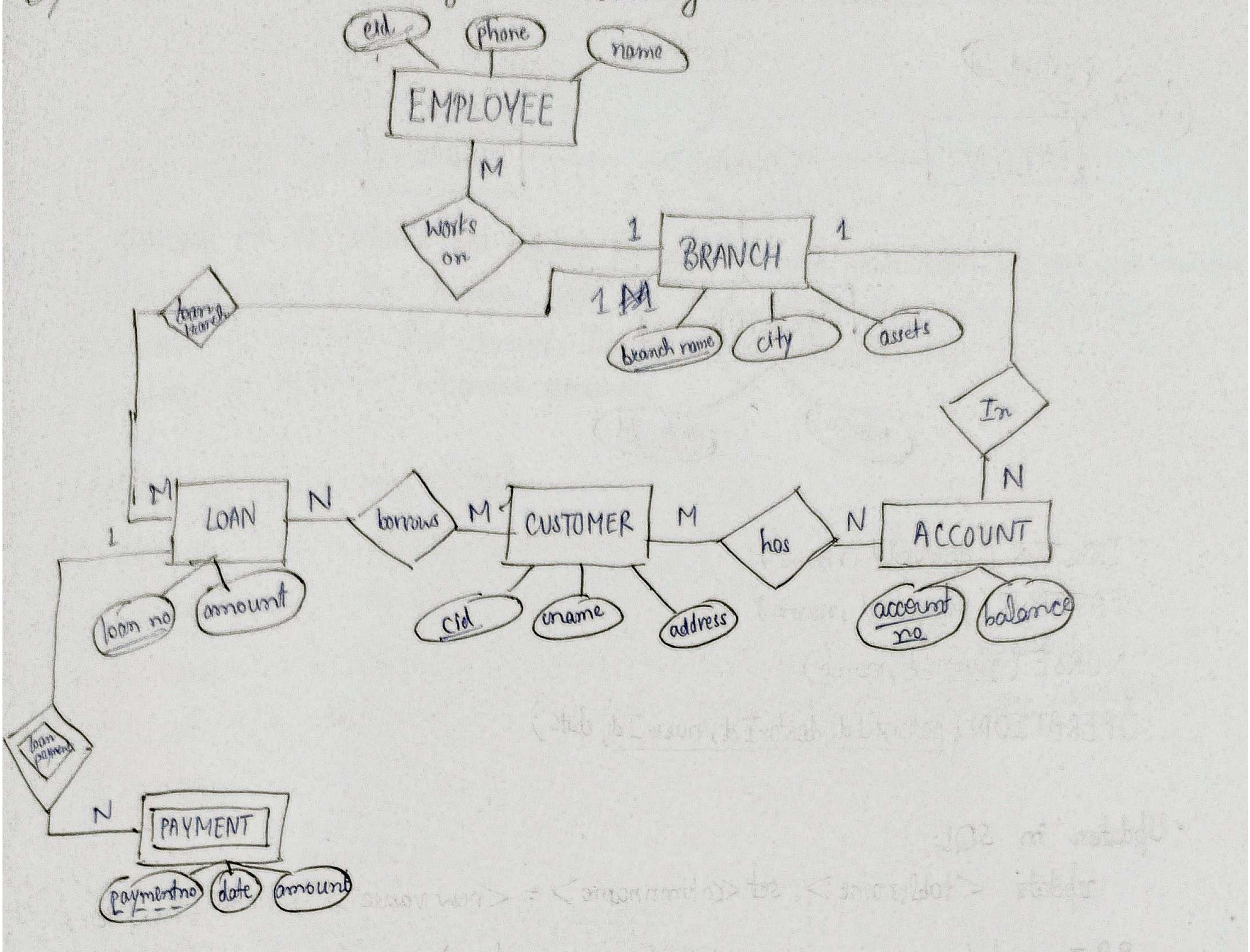
BORROW (cid, loan no)

DEPOSIT ~~HAS~~ (cid, account no)

or

CUSTOMER – ACCOUNT

Q.) Draw relational model for the following E-R model



EMPLOYEE (eid, phone, name, branch name)

BRANCH (branch name, city, assets) ~~loan no~~

ACCOUNT (account no, balance, branch name)

CUSTOMER (cid, cname, address)

LOAN (loan no, amount, branch name)

PAYMENT (payment no, loan no, date, amount)

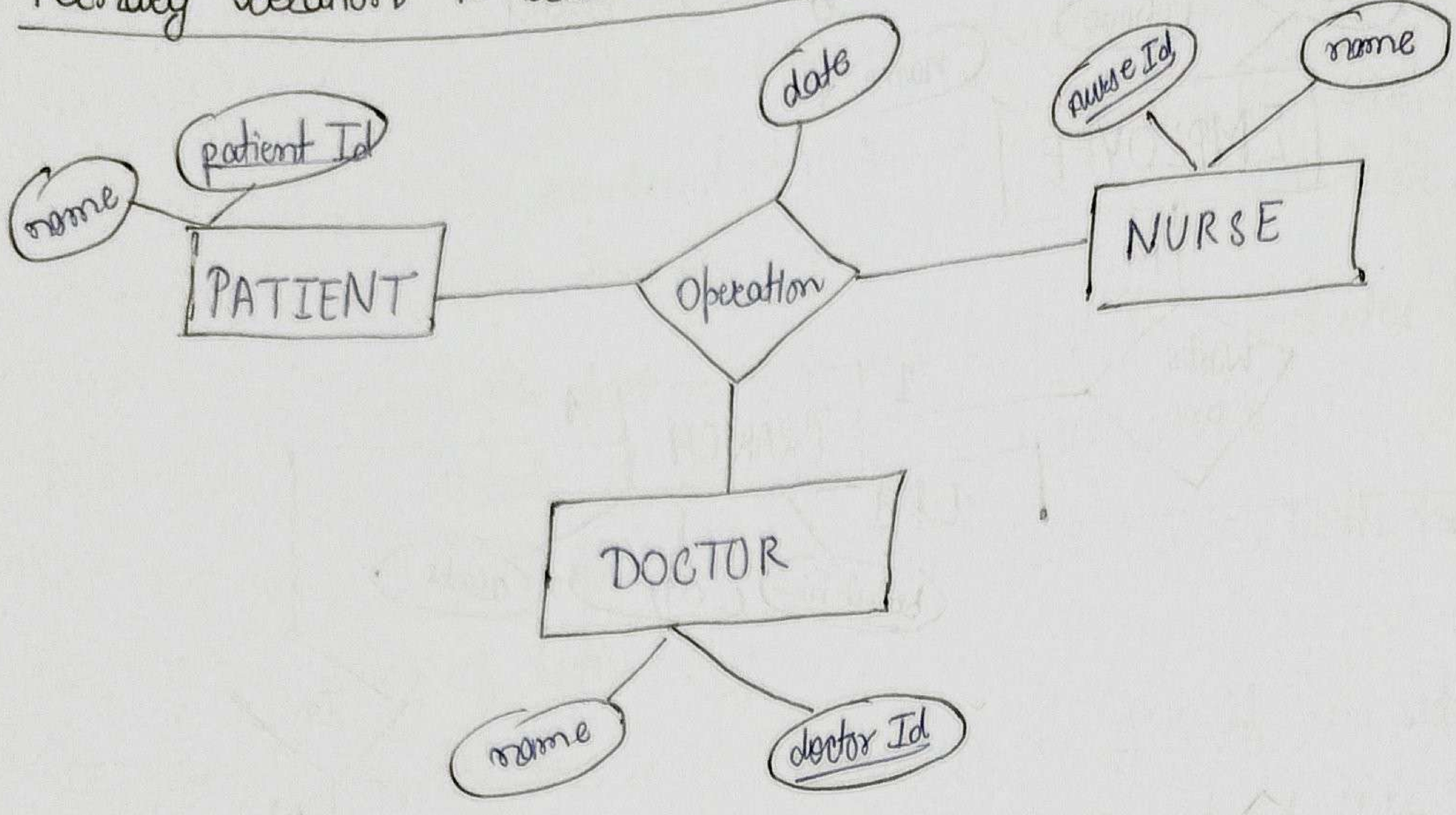
BORROW (cid, loan no)

DEPOSIT ~~HAS~~ (cid, account no)

or

CUSTOMER – ACCOUNT

Ternary relation to relational model



DOCTOR (doctorId , name)

PATIENT (patientId , name)

NURSE (nurseId , name)

OPERATION (patientId, doctorId, nurseId, date)

- Updation in SQL:

  update <tablename> set <columnname> = <new value> where <condition

  e.g.- emp (eno, name, address, dob, designation, salary);

  update emp set address = "Delhi" where address = "Patna";

  update emp set salary = 60000 where salary > 10000 and address = "Mum

- Deletion in SQL:

  delete from <tablename> where <condition>

  e.g.- delete from emp where salary < 10000 and address = "Patna";

- Adding new column in SQL:

  update <tablename> add (<new column name> = datatype (size));

  e.g.- updata emp add (doj date);

- Select in SQL:

  Select <columnname1, columnname 2> from <table name> where <condition>;

  eg- Select (*) from emp order by salary desc;

  (*) → all columns

  order by → shows in ascending

  desc → shows in descending

Professor (PID, pname, dept)

Course (code, dept Id, c name, syllabus)

Prof_course (PID, code, semester)

Q) Identify PK and FK for schema and populate this relation with some tuples and give some examples of insertion operation in the Professor_post course rel^n that violates the referential integrity constraints, and of another that doesn't. Identify various candidate keys for this schema. Make necessary assumptions wherever necessary.

CUSTOMER (custno, cust_name, address)

ORDER (orderno, orderdate, custno, city, amount)

PRODUCT (prodno, price, order no)

Q) Specify FK constraints for above DB. Also insert some tuples in the relation. Show some e.g. of deletion of tuples that violates referential integrity constraints. Make any assumptions wherever necessary.

Relational Algebra

- Formal language for representing relational model

(i) Unary operator — σ, π

(ii) Binary operator — $\cup, \cap, X, -, \div, \bowtie$

σ (Select Operator)

Syntax: $\sigma_{\text{<condition>}}(\text{<tablename>})$

e.g.- emp (ename, id, sal, dno, address)

$\sigma_{\text{address = "Patna"}}(\text{emp})$

↑ equal to

$\sigma_{\text{sal <> 10000}}(\text{emp})$

↑ not equal to

$\sigma_{\text{address = "Patna"} \wedge \text{sal} \geq 10000}(\text{emp})$

↑ and

$\sigma_{\text{address = "Patna"} \vee \text{"Delhi"}}(\text{emp})$

↑ or

* for retrieval of tuples